

Multiple Choice

1. **Answer:** A

Options: A) 4-5 day basis; B) 3-4 day basis; C) 5-6 day basis; D) Hourly basis; E) Monthly basis

Note: The correct answer is 4-5 days because energy balance is typically assessed over several days to account for daily fluctuations in energy intake and expenditure, allowing for a more accurate representation of an individual's overall energy equilibrium.

2. **Answer:** C

Options: A) Have a shorter life expectancy; B) Be less likely to seek medical advice; C) Take fewer medications; D) Pay far less for pharmaceuticals; E) Have higher medical insurance premiums

Note: Identifying links between various diseases can lead to more effective treatment strategies that address multiple conditions simultaneously, potentially reducing the need for older adults to take numerous medications.

3. **Answer:** E

Options: A) Synthesis of fatty acids from glucose; B) Conversion of oxaloacetate to phosphoenolpyruvate; C) Carboxylation of pyruvate to oxaloacetate; D) Conversion of fructose to glucose; E) Synthesis of TMP (thymidine monophosphate)

Note: Folate is essential for the synthesis of TMP, as it serves as a cofactor in the conversion of deoxyuridine monophosphate (dUMP) to TMP, a critical step in DNA synthesis and cell division.

4. **Answer:** A

Options: A) Possesses only RNA; B) A stable and unchanging genome; C) Possesses only DNA; D) Has a single, unsegmented genome; E) Possesses RNA and DNA

Note: The correct answer is accurate because both influenza and HIV are classified as RNA viruses, meaning their genetic material is composed solely of RNA rather than DNA.

5. **Answer:** B

Options: A) Non-seroconverters; B) Seroconverters; C) Short-term survivors with low levels of CD 4+ cells; D) Long-term survivors; E) Long-term survivors with high levels of CD 4+ cells

Note: Seroconverters are individuals who have recently become infected with HIV and have detectable antibodies, allowing researchers to study the progression of the disease from initial infection through various stages, thereby providing a comprehensive understanding of the natural history of HIV/AIDS.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Urea is the primary end product of protein metabolism in humans and is synthesized in the liver, resulting in it comprising approximately 90% of the nitrogen content in urine.

7. **Answer:** True

Options: A) True; B) False

Note: Water is recommended as a safe and effective option for cleaning the mouth because it helps to remove food particles, bacteria, and debris without introducing harmful chemicals or additives that can compromise oral health.

8. **Answer:** True

Options: A) True; B) False

Note: Mutations causing achondroplasia are classified as recessive negative because they lead to a dysfunctional protein that not only affects the individual with the mutation but also impairs the function of the normal allele, resulting in a dominant negative effect.

9. **Answer:** False

Options: A) True; B) False

Note: An oxygen consumption rate of 2 l/min corresponds to an energy expenditure of approximately 12-14 kJ/min, not 200 kJ/min, indicating that the statement is false.

10. **Answer:** False

Options: A) True; B) False

Note: Research indicates that while many individuals experience grief, only a minority—approximately 10-15%—require professional assistance to cope with their loss, making the statement inaccurate.

Long Form Response

11. **Answer:** Papillary muscles play a crucial role in preventing the eversion of the right atrioventricular valve, also known as the tricuspid valve, during ventricular contraction. These muscles are connected to the valve cusps via chordae tendineae, which are fibrous cords that anchor the valve to the ventricular wall. When the right ventricle contracts, the papillary muscles contract as well, pulling on the chordae tendineae to keep the valve leaflets closed and prevent them from inverting into the atrium. This mechanism is vital for maintaining unidirectional blood flow from the right atrium to the right ventricle and ultimately to the lungs for oxygenation. Any dysfunction in the papillary muscles or chordae tendineae can lead to valvular insufficiency, resulting in compromised cardiac function and potential heart failure. Thus, the integrity and performance of the papillary muscles are essential for effective cardiac dynamics and overall cardiovascular health.

Note: correct because it accurately describes the mechanism by which papillary muscles, through their contraction and connection to the tricuspid valve via chordae tendineae, prevent valve eversion during ventricular contraction, thereby ensuring proper cardiac function.

12. **Answer:** The World Health Organization (WHO) recommends zinc supplementation as a crucial intervention for specific populations at risk of deficiency, particularly emphasizing men with high physical activity levels. These individuals often experience increased zinc loss through sweat and higher metabolic demands, which can lead to inadequate dietary intake to meet their needs. Supplementation can thus help mitigate the risk of deficiency and support immune function, muscle recovery, and overall physical performance. Additionally, the WHO highlights the importance of dietary sources of zinc, yet acknowledges that supplementation may be necessary for those engaged in intense physical training or competition. By targeting this population, the recommendations aim to enhance health outcomes and athletic performance.

Note: correct because it accurately identifies the WHO's focus on zinc supplementation for populations at risk, particularly men with high physical activity levels who may face increased zinc loss and metabolic demands, thereby highlighting the importance of supplementation for maintaining adequate zinc status and supporting overall health and performance.